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Telecommunications Adoption and Stock Market Efficiency in Emerging Markets

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Abstract:

Telecommunications has improved price discovery in commodity markets across the developing world. Whether it similarly affects securities markets remains untested. We examine the relationship between telecommunications adoption and stock market efficiency across 20 emerging markets, using runs tests, augmented Dickey-Fuller tests, and variance ratio tests. Efficiency measures based on serial correlation show a positive association between telecommunications adoption and market efficiency, whereas variance ratio tests do not. The association is stronger for telephone and mobile cellular subscriptions than for broadband, which shows no significant relationship with efficiency. Drawing on affordance theory, we argue that telephony affords synchronous communication that supports rapid price discovery, while broadband affords asynchronous information retrieval less directly connected to daily price formation. These findings extend ICT4D research on telecommunications and financial development from banking and mobile money into securities markets, and suggest that basic connectivity may matter more than advanced access for how stock prices incorporate information.

Keywords: Information Technology, Telecommunications Adoption, Stock Market Efficiency, Market Efficiency, Emerging Markets, Financial Development.

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1 Introduction

Telecommunications has transformed how markets function in developing countries. Mobile phones enabled Indian fishermen to check prices across landing sites before choosing where to sell, virtually eliminating waste and raising profits (Jensen, 2007). In Niger, mobile phone coverage substantially reduced grain price dispersion across markets (Aker, 2010). These studies demonstrate a clear mechanism: telecommunications reduces information costs, enabling market participants to act on better information, which improves how accurately prices reflect supply and demand. Yet this evidence comes entirely from commodity markets. Whether telecommunications similarly affects financial markets in developing countries remains unexplored.

Stock markets matter for development because they allocate capital across firms and sectors. When prices accurately reflect available information, resources flow toward productive uses, entrepreneurs can raise funds without collateral, and households can participate in wealth creation. When information frictions distort prices, capital is misallocated, reducing growth and widening inequality (Wurgler, 2000; Levine & Zervos, 1998).

The mechanism linking telecommunications to better price discovery should extend naturally from commodity markets to securities markets. Both involve geographically dispersed participants trading on the basis of information, and both function more efficiently when information costs fall. Telecommunications expansion across Africa, Asia, and Latin America over the past two decades (Lee et al., 2012; Levendis & Lee, 2013; Aker & Mbiti, 2010) has coincided with the growth of securities markets in these regions, yet no study has tested whether telecommunications adoption affects how efficiently stock prices incorporate information. The details of the relationship may differ from commodity markets. Securities markets involve more heterogeneous participants, more complex information, and more elaborate institutional infrastructure. Whether the same basic technologies that improved price discovery for fish and grain also matter for equities, or whether more advanced technologies are required, is an open question. The answer has direct implications for how developing countries prioritize telecommunications investment.

This gap spans two literatures. Finance scholars study market efficiency determinants including liquidity, short-selling constraints, and insider trading regulations (Chordia et al., 2008; Saffi & Sigurdsson, 2011; Fernandes & Ferreira, 2009), but rarely consider telecommunications adoption as an explanatory variable. On the ICT4D side, research on telecommunications and finance has concentrated on banking access, mobile money, and financial inclusion (Chatterjee, 2020; Wamba-Taguimdje & Kala Kamdjoug, 2025; Kpodar & Andrianaivo, 2011). Securities markets are essentially absent. A recent panel study of African stock exchanges finds that mobile subscriptions predict trading volume and fixed telephones predict market capitalization (Igwire & Sibindi, 2022), but this examines market size rather than informational efficiency. The one cross-country study of telecommunications and market efficiency, Lee et al. (2019), pools wealthy and developing nations without adequately controlling for GDP per capita, making it difficult to distinguish telecommunications effects from wealth effects. It also does not engage ICT4D theoretical frameworks. The question of how telecommunications adoption relates to stock market efficiency in developing countries remains open. This paper asks whether different types of telecommunications adoption show different relationships with market efficiency, and what those differences imply for development policy.

We examine this question using data from 20 emerging market countries across Africa, Asia, Latin America, and Eastern Europe. We measure telecommunications adoption through three technologies that represent different affordance structures (Sein et al., 2019): fixed telephone subscriptions, which capture basic voice connectivity; mobile cellular subscriptions, which add geographic flexibility; and fixed broadband subscriptions, which enable data-intensive information retrieval. We assess market efficiency using three complementary statistical tests that capture different dimensions of price behavior: runs tests, augmented Dickey-Fuller tests, and variance ratio tests.

Our findings reveal a consistent pattern across two of three testing approaches. Efficiency measures based on serial correlation (runs tests and augmented Dickey-Fuller tests) show positive associations between telecommunications adoption and market efficiency. Variance ratio tests do not. More notably, the association is strongest for fixed telephone and mobile cellular subscriptions and absent for broadband. Basic voice connectivity, not high-speed data access, tracks with more efficient price formation.

We interpret these results through the lens of affordance theory (Sein et al., 2019; Hatakka et al., 2019). Fixed and mobile telephony afford synchronous, direct communication: traders can negotiate, share

observations, and respond to price movements in real time. Broadband affords asynchronous information retrieval: accessing news, financial statements, or historical data. This distinction speaks to ICT4D debates on appropriate technology and leapfrogging (Davison et al., 2000; Steinmueller, 2001). Rather than assuming more advanced technologies produce superior development outcomes, our evidence suggests that universal access to reliable voice communication may matter more for how prices incorporate information than initiatives focused on high-speed data access. This extends the commodity market evidence into a new domain and reinforces that the relationship between technology and development depends on what a technology enables people to do, not on how advanced it is.

2 Literature Review

Financial markets play a crucial role in economic development. Countries with more developed financial systems grow faster, and the relationship appears causal: exogenous improvements in financial development lead to higher subsequent growth rates (Levine et al., 2000; Beck et al., 2000). The mechanism runs through capital allocation. More developed financial markets channel investment toward growth sectors and away from declining industries (Wurgler, 2000). Stock markets contribute independently of banking: market liquidity robustly predicts long-run growth even after controlling for banking sector development (Levine & Zervos, 1998; Beck & Levine, 2004). This distinction matters for ICT4D research, which has studied telecommunications effects on banking extensively but has largely overlooked securities markets. If telecommunications affects how efficiently stock prices incorporate information, it may affect capital allocation and growth through a channel that existing research has not examined.

The strongest evidence that telecommunications improves market efficiency comes from commodity markets. Jensen (2007) studied the introduction of mobile phone coverage along the Kerala coast in India and found that fishermen used phones to check prices at multiple landing sites before choosing where to sell. Price dispersion across markets fell dramatically, waste was virtually eliminated, and producer profits rose by eight percent while consumer prices fell by four percent. Aker (2010) found similar effects in Niger, where mobile phone coverage reduced grain price dispersion by 10 to 16 percent, with larger effects for more remote market pairs. Aker and Mbiti (2010) synthesize the broader evidence from sub-Saharan Africa, concluding that mobile phones reduce search costs and transaction costs, improving market efficiency across a range of agricultural commodities. The mechanism in each case is the same: telecommunications lowers the cost of acquiring price information, enabling better-informed trading decisions that push prices toward competitive equilibria. This mechanism should not be limited to commodity markets. Securities markets similarly depend on information flowing to participants who then trade on it. Yet no study has tested whether the telecommunications technologies that improved commodity price discovery also affect how efficiently stock prices incorporate information.

Establishing whether telecommunications affects market efficiency requires distinguishing informational efficiency from broader concepts of financial development. A large or deep financial sector is not necessarily one in which prices accurately reflect available information (Levine, 1997; Greenwood & Jovanovic, 1990). A substantial body of research examines how telecommunications affects financial sector development and economic growth in developing countries. Telecommunications infrastructure and financial development jointly predict long-run growth (Pradhan et al., 2015, 2016), and the interaction between telecommunications and finance appears to amplify these growth effects (Sassi & Goaied, 2013; Das et al., 2018). However, these studies measure aggregate financial development through indicators such as banking depth and stock market capitalization rather than informational efficiency. They establish that telecommunications and financial sectors co-evolve, but they do not address whether telecommunications helps prices incorporate information more accurately.

ICT4D research on finance has concentrated heavily on financial inclusion and mobile money. Telecommunications development increases financial inclusion, which also fosters per capita growth (Chatterjee, 2020; Kpodar & Andrianaivo, 2011). Digital finance applications expand financial accessibility while reducing poverty and inequality (Das & Chatterjee, 2023). A recent systematic review confirms that mobile money research treats financial inclusion as the primary channel linking telecommunications to development outcomes (Wamba-Taguimdje & Kala Kamdjoug, 2025).

A small literature examines telecommunications and stock markets directly, but none addresses informational efficiency. Igwilo and Sibindi (2022) study 11 African exchanges and find that mobile phone subscriptions predict trading volume while fixed telephone subscriptions predict market capitalization. Their study measures market development rather than how accurately prices reflect information. Adamolekun et

al. (2023) examine the adoption of electronic trading platforms across 18 African exchanges and find that technology adoption increases listed firms but decreases trading activity, suggesting that technology alone does not automatically improve market functioning. The most directly comparable study to ours is Lee et al. (2019), who find that telecommunications use correlates with stock market efficiency across a large sample of countries. However, their sample pools advanced economies (approximately 37 percent of countries, including Switzerland, Norway, and Denmark) with developing nations. Since wealthier countries have both better telecommunications and more efficient markets, it is difficult for Lee et al. to determine whether telecommunications drives efficiency or whether both reflect wealth. Their data also ends in 2014, before smartphone-based trading matured in emerging markets. We address these limitations by focusing exclusively on emerging markets, properly controlling for GDP per capita, extending the data through 2023, and employing runs tests alongside unit root and variance ratio tests.

3 Theoretical Framework

The Efficient Markets Hypothesis (EMH) posits that financial markets rapidly incorporate available information into asset prices (Fama, 1970). When markets process information efficiently, current prices reflect all known data. Only genuinely new information causes price movements. Since new information cannot be predicted in advance, price changes must be unpredictable, manifesting statistically as random walks (Samuelson, 1965). Information costs are central to this theory. Grossman and Stiglitz (1980) showed that perfectly efficient markets are impossible when information is costly to acquire: there must be sufficient return to compensate informed traders for their information costs, otherwise no one would invest in acquiring information and prices would cease to be informative. Telecommunications reduces these costs. When market participants can acquire and share information more cheaply, more of them find it worthwhile to become informed, and prices reflect available information more accurately. Efficient prices in turn create a feedback loop with the real economy: improvements in how prices incorporate information enhance capital allocation and economic outcomes (Goldstein, 2023).

The EMH in its original formulation treats efficiency as a fixed property of markets. Lo (2004) and Lo (2005) relax this assumption. Lo's Adaptive Markets Hypothesis (AMH) recognizes that efficiency exists on a continuum that evolves as market conditions, institutions, and technologies change. As Lim and Brooks (2010, p. 71) observe, "at the macro level it is not unreasonable to expect market efficiency to evolve over time due to changes in macro institutions, market regulations and information technologies." This framework motivates our empirical approach. If efficiency evolves with technological conditions, then the rapid expansion of telecommunications across emerging markets over the past two decades should be associated with measurable changes in how efficiently stock prices incorporate information.

The EMH and AMH establish that information costs affect efficiency and that efficiency evolves with technological conditions. They do not, however, explain why different telecommunications technologies might show different relationships with efficiency. For this we draw on affordance theory, which has emerged as a central framework in ICT4D for understanding how technologies produce development outcomes (Sein et al., 2019; Hatakka et al., 2019). Affordances are the actions that a technology enables for particular actors in particular contexts. Different technologies afford different actions, and the same technology may afford different actions in different settings.

Applied to telecommunications and market efficiency, affordance theory generates specific expectations. Fixed and mobile telephony afford synchronous, two-way communication. A trader can call a broker, confirm a rumor with a colleague, or negotiate a price in real time. These affordances directly support the kind of rapid information exchange that the EMH identifies as the foundation of efficient pricing. Broadband affords something different: access to large volumes of stored information such as financial statements, news archives, and historical price data. This is valuable, but it supports a different kind of market activity. Retrieving and analyzing stored information helps with fundamental valuation over longer horizons. It is less directly connected to the rapid, interactive information exchange that drives daily price formation. Affordance theory therefore predicts that telephony and broadband should show different relationships with market efficiency, and that the difference should depend on which dimension of efficiency is measured.

Affordance theory explains why different technologies might show different relationships with efficiency. It does not, however, guarantee that any technology will improve efficiency. ICT4D scholarship warns against technological determinism (Toyama, 2011; Avgerou, 2010). Toyama's amplification theory posits that technology magnifies existing institutional forces rather than creating transformation independently. Telecommunications amplifies good governance where it exists and corruption where it prevails. Applied to

capital markets, this suggests that telecommunications can enable efficiency improvements, but cannot substitute for regulatory oversight, investor protections, and market governance.

These frameworks generate testable expectations. Telecommunications adoption should associate with greater market efficiency, and telephony and broadband should show different relationships with efficiency, reflecting their different affordances. In order to test these expectations, we now turn to the data.

4 Empirical Analysis

“So it is not just information that matters, but how widely diffused the information is” (Malkiel et al., 2005, p. 13). Malkiel’s insight captures precisely why telecommunications may enhance market efficiency: by enabling broader information dissemination among market participants.

We restrict our attention to country-level indices in lower-income and middle-income countries. Data on telecommunications adoption are reported yearly. Equity prices, on the other hand, are daily. Financial data are daily, from 1st January, 1991 to 31 Dec. 2023. The daily price data are sourced from Yahoo Finance covering 20 emerging market countries across four regions: Africa, Asia, Latin America, and Eastern Europe. Returns are calculated as the log difference of the adjusted closing prices.

Table 1. Emerging Market ETFs and Indices

Country	ETF	Index
Argentina	ARGT	MERV
Brazil	EWZ	BVSP
Chile	ECH	IPSA
Colombia	GXG	–
Czech Rep.	–	PX
Egypt	EGPT	–
Hungary	–	BUX
India	EPI, INDY, INDA	BSESN, NSEI
Indonesia	EIDO	IDX, JKSE
Malaysia	EWM	KLSE
Mexico	EWX	MXX
Nigeria	NGE	–
Pakistan	PAK	KSE
Peru	EPU	SPBLPGPT
Philippines	EPHI	PSEI_PS
Poland	EPOL	–
S. Africa	EZA	J203_JO
Thailand	–	SET_BK
Turkey	TUR	XU100
Vietnam	VNM	–

Our dataset includes both country-specific ETFs and local stock exchange indices, with indicator variables distinguishing between the two instrument types. This dual approach provides robustness by capturing market efficiency through different investment vehicles. Table 1 presents the complete list of emerging market ETFs and stock exchange indices used in our analysis.

Based on our theoretical framework, we expect two patterns. First, telecommunications adoption should associate positively with market efficiency. Second, technologies affording synchronous communication (telephony) should show stronger associations than those affording asynchronous information retrieval (broadband).

Single-method studies may provide an incomplete and inconsistent picture of efficiency in developing markets. Multiple testing approaches therefore prove valuable (Lim, 2009; Worthington & Higgs, 2005). We

examine the relationship between telecommunications and market efficiency using three complementary methodologies that test different aspects of the random walk hypothesis: (1) runs tests, (2) Augmented Dickey-Fuller tests, and (3) variance ratio tests. Each method relies on different assumptions and tests different departures from random walks. After extracting the appropriate efficiency measures (details below), we regress each measure on the telecommunications variables, including and excluding controls for lnGDP and a dummy variable for country-wide ETFs.

We use three different measures of telecommunications adoption, all collected by the International Telecommunications Union: (1) fixed broadband subscriptions per 100 people, (2) mobile cellular subscriptions per 100 people, and (3) total telephone subscriptions (cellular plus fixed-line) per 100 people. The data are yearly, by country, from 1991 to 2024.

The practice of regressing market efficiency on institutional and technological variables is well-established. Stock market efficiency correlates with institutional factors such as legal protections (Fernandes & Ferreira, 2009; Lim & Brooks, 2010) and disclosure requirements (Jin & Myers, 2006). It also responds to market microstructure variables such as liquidity and trading volume (Chung & Hrazdil, 2010; Griffin et al., 2010). Finally, technological changes affect how efficiently markets process information.¹ These studies establish precedents for treating efficiency as an outcome variable, supporting our approach.

4.1 Runs Tests

Runs tests are a non-parametric test for efficiency. A run is a sequence of consecutive returns sharing the same sign. For example, three consecutive positive returns constitute a single run of length three.

Runs tests have several advantages over competing methods. Since only the sign matters, researchers need not make distributional assumptions about the returns process. They are merely counting sequences rather than relying on normality assumptions. Since the sign but not the magnitude of the change is tracked, the measure is robust to outliers. Finally, runs tests do not rely on large-sample or asymptotic approximations; the finite sample distribution is known analytically.

A runs test compares the observed number of runs with the expected number of runs under the null hypothesis of randomness. Specifically,

$$Z_{i,t} = (R_{i,t} - E(R_{i,t})) / \sigma_{i,t}$$

where $R_{i,t}$ is the observed number of runs (both positive and negative) in country i during year t . The expected number of runs under pure randomness is

$$E(R_{i,t}) = 1 + (2n_{i,t}^{(+)}n_{i,t}^{(-)})/n_{i,t}$$

where the number of positive runs is denoted $n_{i,t}^{(+)}$, the number of negative runs is $n_{i,t}^{(-)}$, and the number of observations is n . Finally,

$$\sigma_{i,t}^2 = [2n_{i,t}^{(+)}n_{i,t}^{(-)}(2n_{i,t}^{(+)}n_{i,t}^{(-)} - n_{i,t})] / [n_{i,t}^2(n_{i,t} - 1)].$$

The closer $Z_{i,t}$ is to zero, the closer to a random walk; the larger in magnitude is $Z_{i,t}$, the more runs and the less efficient is the underlying series.

To aid in interpretation, we construct the following efficiency statistic:

$$E_{i,t}^{Runs} = 1 - |Z_{i,t}|$$

Thus, higher values of $E_{i,t}^{Runs}$ indicate higher efficiency.

These efficiency measures are calculated for the calendar year for each index/EFT.

¹ Gu and Finnerty (2002) document that U.S. market efficiency increased markedly after the technological advances of the 1980s. Naidu and Rozeff (1994) find that Singapore's automated trading system improved efficiency whereas Dicle and Levendis (2013) find that Johannesburg's system did not.

To test whether telecommunications adoption is correlated with greater efficiency, we estimate an OLS regression of the form:

$$E_{i,t}^{Runs} = \beta_0 + \beta_1 IT_{c,t} + \beta_2 \ln(GDP)_{c,t} + \beta_3 ETF_i + \epsilon_{i,t}$$

and a fixed effects regression of the form:

$$E_{i,t}^{Runs} = \alpha_i + \beta_1 IT_{c,t} + \beta_2 \ln(GDP)_{c,t} + \epsilon_{i,t}$$

Table 2. Runs Test OLS Results

	(1)	(2)	(3)	(4)	(5)	(6)
	BBND	CEL	TEL	BBND	CEL	TEL
Broadband	0.002			0.004		
	(0.541)			(0.362)		
Cellular		0.001			0.001*	
		(0.102)			(0.085)	
Telephones			0.001*			0.001*
			(0.095)			(0.092)
lnGDP				-0.029	0.023	0.012
				(0.590)	(0.579)	(0.782)
ETF				-0.106*	-0.079	-0.078
				(0.059)	(0.169)	(0.174)
Constant	0.049	-0.059	-0.073	0.336	-0.226	-0.153
	(0.228)	(0.368)	(0.314)	(0.457)	(0.513)	(0.667)
Observations	650	706	706	650	706	706

Table. 3 Runs Test FE Results

	(1)	(2)	(3)	(4)	(5)	(6)
	BBND	CEL	TEL	BBND	CEL	TEL
Broadband	0.007			-0.003		
	(0.191)			(0.648)		
Cellular		0.002**			-0.001	
		(0.032)			(0.374)	
Telephones			0.002**			-0.001
			(0.033)			(0.541)
lnGDP				0.465***	0.759***	0.704***
				(0.003)	(0.001)	(0.001)
Constant	0.015	-0.139*	-0.165*	-3.904***	-6.353***	-5.904***
	(0.684)	(0.076)	(0.068)	(0.003)	(0.001)	(0.001)
Observations	650	706	706	650	706	706

Runs tests demonstrate a consistent pattern: telephone use shows the strongest association with efficiency, followed by cellular subscriptions, with broadband showing no relationship. This is consistent with the affordance-based expectation that technologies enabling synchronous communication show stronger associations with efficiency than those enabling asynchronous information retrieval. The relationship is also notably stronger in the simple bivariate specifications compared to those with GDP and ETF controls, indicating that part of the telecommunications-efficiency relationship operates through general economic development.

The inclusion of runs tests complements the unit root and variance ratio approaches employed by Lee et al. (2019) which we turn to shortly. While unit root tests examine stationarity and variance ratio tests examine variance scaling properties, runs tests capture sequential randomness in return patterns. Finding consistent results across these three distinct dimensions of market efficiency strengthens confidence that observed relationships represent genuine regularities rather than artifacts of particular testing methodologies.

4.2 Augmented Dickey-Fuller Tests

For each ticker i , we calculate daily returns as the log of first-differences in daily prices:

$$Ret_{i,d} = \ln(P_{i,d}) - \ln(P_{i,d-1})$$

where d indexes trading days within year t .

We then run an Augmented Dickey-Fuller test with drift and two lags on the daily returns for each ticker-year:

$$Ret_{i,d} = \beta_{i,0} + \beta_{i,1}Ret_{i,d-1} + \beta_{i,2}Ret_{i,d-2} + \epsilon_{i,d}$$

The key statistic from this regression is the ADF test statistic $Z_{i,t}$. More negative values indicate stronger rejection of the unit root hypothesis. Preferring higher numbers to indicate more efficiency, we construct:

$$E_{i,t}^{ADF} = -Z_{i,t}$$

Finally, to test whether our ADF-based efficiency statistic is correlated with telecommunications adoption, we estimate three different types of regressions: (1) pooled OLS,

$$E_{i,t}^{ADF} = \alpha + \beta_1 IT_{i,t} + \beta_2 \ln(GDP)_{i,t} + \beta_3 ETF_i + \epsilon_{i,t}$$

(2) panel fixed-effects

$$E_{i,t}^{ADF} = \alpha_i + \beta_1 IT_{i,t} + \beta_2 \ln(GDP)_{i,t} + \epsilon_{i,t}$$

where the time-invariant ETF variable is absorbed by the individual (time-invariant) fixed effect, α_i ; and (3) a Random effects model:

$$E_{i,t}^{ADF} = \alpha + \beta_1 IT_{i,t} + \beta_2 \ln(GDP)_{i,t} + \beta_3 ETF_i + u_i + \epsilon_{i,t}.$$

Table 4. ADF Test OLS Results

	(1)	(2)	(3)	(4)	(5)	(6)
	BBND	CEL	TEL	BBND	CEL	TEL
Broadband	0.006			-0.014		
	(0.520)			(0.297)		
Cellular		0.005***			0.001	
		(0.000)			(0.408)	
Telephones			0.005***			0.001
			(0.000)			(0.440)
lnGDP				0.256**	0.211**	0.202**
				(0.021)	(0.015)	(0.037)
ETF				0.521***	0.513***	0.518***
				(0.000)	(0.000)	(0.000)
Constant	4.050***	3.551***	3.486***	1.759**	1.877***	1.936***
	(0.000)	(0.000)	(0.000)	(0.047)	(0.005)	(0.007)
Observations	650	706	706	650	706	706

Table 5. ADF Test FE Results

	(1)	(2)	(3)	(4)	(5)	(6)
	BBND	CEL	TEL	BBND	CEL	TEL
Broadband	0.003			-0.011		
	(0.752)			(0.305)		
Cellular		0.003*			-0.003	
		(0.067)			(0.239)	
Telephones			0.003*			-0.003
			(0.075)			(0.235)
lnGDP				0.327*	0.689**	0.692**
				(0.081)	(0.024)	(0.023)
ETF				0.000	0.000	0.000
				(.)	(.)	(.)
Constant	4.074***	3.763***	3.737***	1.384	-1.517	-1.502
	(0.000)	(0.000)	(0.000)	(0.368)	(0.506)	(0.504)
Observations	650	706	706	650	706	706

Table 6. ADF Test RE Results

	(1)	(2)	(3)	(4)	(5)	(6)
	BBND	CEL	TEL	BBND	CEL	TEL
Broadband	0.005			-0.014		
	(0.592)			(0.148)		
Cellular		0.005***			0.001	
		(0.006)			(0.426)	
Telephones			0.005***			0.001
			(0.005)			(0.462)
lnGDP				0.256***	0.211**	0.202*
				(0.006)	(0.029)	(0.060)
ETF				0.521***	0.513***	0.518***
				(0.000)	(0.000)	(0.000)
Constant	4.057***	3.597***	3.542***	1.759**	1.877**	1.936**
	(0.000)	(0.000)	(0.000)	(0.020)	(0.015)	(0.019)
Observations	650	706	706	650	706	706

The ADF results largely mirror the runs test findings, with basic telephone use more correlated than cellular. The consistent significance of telephone and cellular in the OLS specifications, but their reduced significance in fixed effects models, suggests that much of the relationship might operate through cross-sectional differences between countries rather than within-country changes over time. Alternatively, this could also be the result of the slow evolution of telecommunications adoption. Relatively stable variables are nearly indistinguishable from constants, which get differenced out in FE estimations.

4.3 Variance Ratio Tests

Introduced by Lo and MacKinlay (1988), VR tests are the most commonly used tests of market efficiency over the last two decades. VR tests compare the variance of percentage returns calculated over two different time spans. Here, we compare the variance of returns calculated daily, with those calculated over five-day

spans (the trading week). Under the assumption of a random walk, the five-day returns should have a variance that is five times greater than the variance for single-day returns.

First, we calculate 1-day and 5-day percentage returns as log differences:

$$Ret_{i,d}^1 = \ln(P_{i,d}) - \ln(P_{i,d-1})$$

and

$$Ret_{i,d}^5 = \ln(P_{i,d}) - \ln(P_{i,d-5})$$

with their corresponding variances denoted as: $V(Ret_i^1)$ and $V(Ret_i^5)$ calculated over the calendar year.

If returns follow a random walk, then $V(Ret_i^5)$ will be five times larger than $V(Ret_i^1)$, so the variance ratio defined as

$$VR_{i,t} = V(Ret_{i,t}^5)/5V(Ret_{i,t}^1)$$

will equal one under the null of a random walk. The larger the departure from a random walk, the less efficient the market, and the larger the VR statistic will be.

Preferring larger numbers to indicate more efficiency, we create the following efficiency statistic which measures deviations in both directions of the predicted ratio:

$$E_{i,t}^{VR} = 1 - |VR_{i,t} - 1|$$

To clarify, our VR statistics are calculated by comparing 1-day and 5-day returns over a 1-year window. Thus, we have yearly VR statistics for each stock market. We do not calculate VRs at higher frequencies for several reasons. First, the telecommunications data is yearly. Inter-day frequencies give ample data. Further, Andersen et al. (2001) has shown that VR estimates are quite noisy at higher frequencies.

Our research question is whether market efficiency is related to telecommunications adoption in a country. To answer this, we estimate the following model

$$E_{i,t}^{VR} = \beta_0 + \beta_1 IT_{i,t} + \beta_2 \ln(GDP)_{i,t} + \beta_3 ETF_i + \epsilon_{i,t}$$

with and without controls. Since the ETF variable is constant across time for each ticker, it is perfectly colinear with the fixed effect and is not identified. Thus, for the FE and RE models it is excluded as a control.

Table 7. VR OLS Results

	(1)	(2)	(3)	(4)	(5)	(6)
	BBND	CEL	TEL	BBND	CEL	TEL
Broadband	-0.118			-0.117		
	(0.319)			(0.331)		
Cellular		-0.003			-0.002	
		(0.234)			(0.341)	
Telephones			-0.004			0.000
			(0.238)			(0.960)
lnGDP				-0.037	-0.791	-0.830
				(0.623)	(0.305)	(0.307)
ETF				1.138	0.987	0.935
				(0.353)	(0.360)	(0.362)
Constant	0.936***	0.383	0.462*	0.667	6.539	6.745
	(0.003)	(0.218)	(0.056)	(0.306)	(0.249)	(0.253)

Table 7. VR OLS Results

Observations	652	711	711	652	711	711
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Table 8. VR FE Results

	(1)	(2)	(3)	(4)	(5)	(6)
	BBND	CEL	TEL	BBND	CEL	TEL
Broadband	-0.254			-0.194		
	(0.331)			(0.358)		
Cellular		-0.011			0.032	
		(0.327)			(0.313)	
Telephones			-0.008			0.036
			(0.325)			(0.304)
lnGDP				-3.071	-11.161	-11.682
				(0.318)	(0.304)	(0.297)
Constant	1.964	1.033	0.896	27.838	92.387	96.086
	(0.321)	(0.278)	(0.269)	(0.312)	(0.303)	(0.296)
Observations	652	711	711	652	711	711

Table 9. VR RE Results

	(1)	(2)	(3)	(4)	(5)	(6)
	BBND	CEL	TEL	BBND	CEL	TEL
Broadband	-0.118			-0.112		
	(0.333)			(0.357)		
Cellular		-0.003			0.002	
		(0.287)			(0.490)	
Telephones			-0.004			0.003
			(0.277)			(0.416)
lnGDP				-0.091	-0.853	-0.918
				(0.594)	(0.303)	(0.307)
Constant	0.936***	0.383	0.462*	1.672	7.202	7.578
	(0.007)	(0.216)	(0.064)	(0.236)	(0.257)	(0.262)
Observations	652	711	711	652	711	711

In contrast to the runs and ADF results, the VR tests show no consistent relationship between telecommunications adoption and efficiency. While telecommunications adoption appears to reduce the patterns of serial correlation that runs and ADF tests detect, it does not systematically affect the variance scaling properties that VR tests measure.

5 Discussion and Policy Implications

The relationship between telecommunications adoption and market efficiency depends on the measure we employ. Runs and ADF tests indicate that higher levels of telecommunications adoption correlate positively with market efficiency. Variance ratio tests, however, do not. This mixed pattern is not entirely unexpected, given Worthington and Higgs (2005)'s finding that different efficiency tests can yield contradictory conclusions, even when applied to the same markets. Different tests capture different dimensions of market behavior and test for different types of departures from random walks using different assumptions.

The divergence between serial correlation tests (runs, ADF) and variance tests (VR) suggests telecommunications affects specific dimensions of efficiency. Telecommunications adoption may eliminate

predictable patterns in returns without necessarily affecting the variance scaling properties that VR tests measure. This heterogeneity is theoretically informative: it suggests telecommunications improves information flow sufficiently to reduce exploitable patterns, but does not fundamentally alter market features that determine variance dynamics at different time scales.

The variance ratio results warrant careful interpretation within the informational environment of emerging markets. Griffin et al. (2010) argue that efficiency measures can be misleading when they ignore the informational context in which markets operate. In emerging markets with limited information flow, VR tests may capture noise trading or liquidity-driven variance patterns rather than information-driven price movements.

Our finding that basic telecommunications technologies show stronger associations with market efficiency than broadband extends a pattern observed in other contexts. In commodity markets, basic mobile telephony drove the efficiency gains that Jensen (2007) and Aker (2010) documented. Igwilo and Sibindi (2022) found that fixed telephone subscriptions, not mobile or internet, predicted stock market capitalization across African exchanges. Our results suggest the same pattern holds for informational efficiency: technologies affording synchronous communication between market participants show stronger relationships with how accurately prices reflect information than technologies affording access to stored data.

Daily price formation depends on rapid, interactive information exchange rather than the retrieval and analysis of documents. The finding also complicates the leapfrogging narrative (Davison et al., 2000; Steinmueller, 2001). In the specific domain of market efficiency, the older technology appears to matter more, not because it is technically superior, but because its affordances better match the informational demands of daily price discovery.

By focusing exclusively on emerging markets and controlling for GDP per capita, our analysis provides more targeted evidence than broad cross-country studies pooling heterogeneous economies. Lee et al. (2019) find stronger associations in a larger sample that includes wealthy nations and does not control for GDP. Since wealthier countries have both better telecommunications and more efficient markets, their results may capture wealth effects rather than telecommunications effects. Our GDP controls and emerging-market-only sample address this confound. That our results are weaker but still present in two of three testing approaches suggests that telecommunications does associate with efficiency independently of income, though the relationship is more modest than pooled samples imply.

These results counsel against overconfidence. The positive correlations we observe in two of three tests suggest efficiency benefits are real but should be viewed as complementary to traditional financial sector development tools rather than the primary justification for telecommunications investment. Amplification theory (Toyama, 2011; Avgerou, 2010) predicts that telecommunications magnifies existing institutional forces. The efficiency-enhancing properties we observe likely require complementary institutions: regulatory oversight, disclosure requirements, and investor protections. The ability to disseminate information easily means little if the information is not accurate or credible.

For development policy, our findings suggest that universal access to reliable voice communication may matter more for capital market efficiency than initiatives focused on high-speed data access. This does not imply broadband is unimportant for development generally, but for the specific function of daily price formation, synchronous communication appears more consequential. More broadly, this study extends ICT4D financial development research beyond its current focus on mobile money, banking, and financial inclusion to securities markets and how prices come to reflect available information. Capital markets represent a significant institution for resource allocation in developing countries, and understanding how telecommunications affects their functioning opens a new area of inquiry for ICT4D scholarship.

6 Limitations and Future Research

This study provides initial evidence on the relationship between telecommunications adoption and financial market efficiency in emerging markets. Our findings suggest several promising directions for future research at the intersection of ICT4D and financial market development.

Future work could examine the effects of telecommunications across different market contexts. Do telecommunications-efficiency relationships vary by market size, trading volume, or institutional quality? Does the basic-versus-advanced technology pattern we observe hold across different regulatory environments?

Our market-level analysis masks potential heterogeneity across firms and sectors. Do telecommunications particularly benefit smaller firms with less analyst coverage, or larger firms with more complex information environments? Sector-level analysis could also reveal whether the effects of telecommunications vary across industries with different information intensities. Future work could leverage regional or demographic differences in telecommunications access to identify more granular effects.

We focus on market efficiency as a development outcome, but telecommunications adoption likely affects other financial market dimensions including liquidity, volatility, trading costs, and market participation rates. Understanding the full range of telecommunications effects would inform more comprehensive cost-benefit analyses of telecommunications policies.

Econometrically, our analysis is correlational and does not establish causation. Our results are strongest in pooled specifications. Country fixed effects reduce the significance of the telecommunications coefficients, and two-way fixed effects with both entity and year effects eliminate them entirely across all three efficiency tests. This reflects a fundamental challenge in studying slow-moving infrastructure variables. Mobile cellular subscriptions correlate 0.81 with calendar year in our sample: telecommunications expanded globally in a near-monotonic trend over the 1991 to 2023 period, leaving little residual variation once year effects are removed.

As a partial check, we averaged the data into five-year periods to allow more within-country variation. Under entity fixed effects with five-year averaging, cellular and telephone subscriptions significantly predict runs-test efficiency ($p = 0.005$ and $p = 0.006$ respectively), suggesting that the relationship is not purely cross-sectional. However, these results do not survive the addition of $\ln\text{GDP}$ as a control or time fixed effects. These specifications are reported in the Appendix as Tables 11, 12 and 13. We interpret our findings as documenting a robust cross-country association between telecommunications adoption and one dimension of market efficiency. Causal identification remains to be established through research designs better suited to slow-moving infrastructure variables, such as event studies exploiting discrete changes in telecommunications regulation.

Future research might account for structural breaks in efficiency or telecommunications series (Perron, 1989; Levendis, 2023). Panel approaches accommodating cross-sectional dependence and structural breaks could provide more robust evidence (Narayan, 2008). Given the international nature of investment capital, cross-country spillovers or regional effects (Bai, 2009; Bai & Ng, 2010; Pesaran et al., 2004) also provide fruitful areas for further research.

Disclosure statement

The author reports no competing interests

References

- Adamolekun, G., Sakariyahu, R., Lawal, R., & Ahmed, A. (2023). Electronic trading and stock market participation in Africa: Does technology induce participation? *Economics Letters*, 224, 110991.
- Aker, J. C. (2010). Information from markets near and far: Mobile phones and agricultural markets in Niger. *American Economic Journal: Applied Economics*, 2(3), 46–59.
- Aker, J. C., & Mbiti, I. M. (2010). Mobile phones and economic development in Africa. *Journal of economic Perspectives*, 24(3), 207–232.
- Andersen, T. G., Bollerslev, T., & Das, A. (2001). Variance-ratio statistics and high-frequency data: Testing for changes in intraday volatility patterns. *The Journal of Finance*, 56(1), 305–327.
- Avgerou, C. (2010). Discourses on ICT and development. *Information Technologies and International Development*, 6(3), 1–18.
- Bai, J. (2009). Panel data models with interactive fixed effects. *Econometrica*, 77(4), 1229–1279.
- Bai, J., & Ng, S. (2010). Panel unit root tests with cross-section dependence: A further investigation. *Econometric Theory*, 26(4), 1088–1114.
- Beck, T., & Levine, R. (2004). Stock markets, banks, and growth: Panel evidence. *Journal of Banking & Finance*, 28(3), 423–442.
- Beck, T., Levine, R., & Loayza, N. (2000). Finance and the sources of growth. *Journal of Financial Economics*, 58(1-2), 261–300.
- Chatterjee, A. (2020). Financial inclusion, information and communication technology diffusion, and economic growth: A panel data analysis. *Information Technology for Development*, 26, 607–635.
- Chordia, T., Roll, R., & Subrahmanyam, A. (2008). Liquidity and market efficiency. *Journal of Financial Economics*, 87(2), 249–268.
- Chung, D., & Hrazdil, K. (2010). Liquidity and market efficiency: A large sample study. *Journal of Banking & Finance*, 34(10), 2346–2357.
- Das, A., Chowdhury, M., & Seaborn, S. (2018). ICT diffusion, financial development and economic growth: New evidence from low and lower middle-income countries. *Journal of the Knowledge Economy*, 9(3), 928–947.
- Das, S., & Chatterjee, A. (2023). Impacts of ICT and digital finance on poverty and income inequality: A sub-national study from India. *Information Technology for Development*, 29, 378–405.
- Davison, R., Vogel, D., Harris, R., & Jones, N. (2000). Technology leapfrogging in developing countries—an inevitable luxury?. *The Electronic Journal of Information Systems in Developing Countries*, 1(1), 1–10.
- Dicle, M. F., & Levendis, J. (2013). The impact of technological improvements on developing financial markets: The case of the Johannesburg stock exchange. *Review of Development Finance*, 3(4), 204–213.
- Fama, E. F. (1970). Efficient capital markets. *Journal of Finance*, 25(2), 383–417.
- Fernandes, N., & Ferreira, M. A. (2009). Insider trading laws and stock price informativeness. *The Review of Financial Studies*, 22(5), 1845–1887.
- Goldstein, I. (2023). Information in financial markets and its real effects. *Review of Finance*, 27(1), 1–32.
- Greenwood, J., & Jovanovic, B. (1990). Financial development, growth, and the distribution of income. *Journal of Political Economy*, 98(5, Part 1), 1076–1107.
- Griffin, J. M., Kelly, P. J., & Nardari, F. (2010). Do market efficiency measures yield correct inferences? A comparison of developed and emerging markets. *The Review of Financial Studies*, 23(8), 3225–3277.
- Grossman, S. J., & Stiglitz, J. E. (1980). On the impossibility of informationally efficient markets. *The American Economic Review*, 70(3), 393–408.

- Gu, A. Y., & Finnerty, J. (2002). The evolution of market efficiency: 103 years daily data of the Dow. *Review of Quantitative Finance and Accounting*, 18, 219–237.
- Hatakka, M., Saebo, O. & Thapa, D. (2019). A framework to explain the relation between ICT and development: Combining affordances and the capability approach. In *International Conference on Social Implications of Computers in Developing Countries* (pp. 60–71). Springer.
- Igwilo, J. I., & Sibindi, A. B. (2022). ICT adoption and stock market development: Empirical evidence using a panel of African countries. *Risks*, 10(2), 25.
- Jensen, R. (2007). The digital provide: Information (technology), market performance, and welfare in the South Indian fisheries sector. *The quarterly journal of economics*, 122(3), 879–924.
- Jin, L., & Myers, S. C. (2006). R-squared around the world: New theory and new tests. *Journal of Financial Economics*, 79(2), 257–292.
- Kpodar, M. K., & Andrianaivo, M. (2011). *ICT, financial inclusion, and growth: Evidence from African countries*. International Monetary Fund.
- Lee, M., Tsai, T., Chen, J., & Lio, M. (2019). Can information and communication technology improve stock market efficiency? a cross-country study. *Bulletin of Economic Research*, 71(2), 113–135.
- Lee, S. H., Levendis, J., & Gutierrez, L. (2012). Telecommunications and economic growth: An empirical analysis of Sub-Saharan Africa. *Applied Economics*, 44(4), 461–469.
- Levendis, J. (2023). *Time series econometrics* (2nd ed.). Springer.
- Levendis, J., & Lee, S. H. (2013). On the endogeneity of telecommunications and economic growth: Evidence from Asia. *Information Technology for Development*, 19, 62–85.
- Levine, R. (1997). Financial development and economic growth: Views and agenda. *Journal of Economic Literature*, 35(2), 688–726.
- Levine, R., & Zervos, S. (1998). Stock markets, banks, and economic growth. *American Economic Review*, 537–558.
- Levine, R., Loayza, N., & Beck, T. (2000). Financial intermediation and growth: Causality and causes. *Journal of Monetary Economics*, 46(1), 31–77.
- Lim, K. P. (2009). *An empirical analysis of the weak-form efficiency of stock markets* [Doctoral dissertation]. Monash University.
- Lim, K., & Brooks, R. D. (2010). Why do emerging stock markets experience more persistent price deviations from a random walk over time? a country-level analysis. *Macroeconomic Dynamics*, 14(S1), 3–41.
- Lo, A. W. (2004). The adaptive markets hypothesis: Market efficiency from an evolutionary perspective. *Journal of Portfolio Management*.
- Lo, A. W. (2005). Reconciling efficient markets with behavioral finance: The adaptive markets hypothesis. *Journal of Investment Consulting*, 7(2), 21–44.
- Lo, A. W., & MacKinlay, A. C. (1988). Stock market prices do not follow random walks: Evidence from a simple specification test. *Review of Financial Studies*, 1(1), 41–66.
- Malkiel, B., Mullainathan, S., & Stangle, B. (2005). Market efficiency versus behavioral finance. *Journal of Applied Corporate Finance*, 17(3), 124–136.
- Naidu, G., & Rozeff, M. S. (1994). Volume, volatility, liquidity and efficiency of the Singapore stock exchange before and after automation. *Pacific-Basin Finance Journal*, 2(1), 23–42.
- Narayan, P. K. (2008). Do shocks to G7 stock prices have a permanent effect? evidence from panel unit root tests with structural change. *Mathematics and Computers in Simulation*, 77(4), 369–373.
- Perron, P. (1989). The great crash, the oil price shock, and the unit root hypothesis. *Econometrica*, 57(6), 1361–1401.
- Pesaran, M. H., Schuermann, T., & Weiner, S. M. (2004). Modeling regional interdependencies using a global error-correcting macroeconometric model. *Journal of Business & Economic Statistics*, 22(2), 129–162.

- Pradhan, R. P., Arvin, M. B., & Hall, J. H. (2016). Economic growth, development of telecommunications infrastructure, and financial development in Asia, 1991–2012. *The Quarterly Review of Economics and Finance*, 59, 25–38.
- Pradhan, R. P., Arvin, M. B., & Norman, N. R. (2015). The dynamics of information and communications technologies infrastructure, economic growth, and financial development: Evidence from Asian countries. *Technology in Society*, 42, 135–149.
- Saffi, P. A., & Sigurdsson, K. (2011). Price efficiency and short selling. *The Review of Financial Studies*, 24(3), 821–852.
- Samuelson, P. (1965). Proof that properly anticipated prices fluctuate randomly. *Industrial Management Review (pre-1986)*, 6(2), 41–49.
- Sassi, S., & Goaied, M. (2013). Financial development, ICT diffusion and economic growth: Lessons from MENA region. *Telecommunications Policy*, 37(4-5), 252–261.
- Sein, M. K., Thapa, D., Hatakka, M., & S{\ae}b{o}, {. (2019). A holistic perspective on the theoretical foundations for ICT4D research. *Information Technology for Development*, 25, 7–25.
- Steinmueller, W. E. (2001). ICTs and the possibilities for leapfrogging by developing countries. *International Labour Review*, 140, 193.
- Toyama, K. (2011). Technology as amplifier in international development. In *Proceedings of the 2011 iConference* (pp. 75–82). .
- Wamba-Taguimdje, S., & Kala Kamdjoug, J. R. (2025). Mobile payments and money technologies in sustainable development: A systematic literature review and computer-assisted interpretive analysis. *Information Technology for Development*, 31, 435–472.
- Worthington, A. C., & Higgs, H. (2005). *Weak-form market efficiency in asian emerging and developed equity markets: Comparative tests of random walk behaviour* (Working Paper 3). School of Accounting & Finance, University of Wollongong.
- Wurgler, J. (2000). Financial markets and the allocation of capital. *Journal of Financial Economics*, 58(1-2), 187–214.

Appendix A: Example of Dataset

The raw financial data was daily, for each country's stock tickers. After calculating efficiency statistics for each ticker, the data set was collapsed to the yearly level so that they could be regressed on the telecommunications data which was yearly, by country. Table A1 shows a few representative rows of the data set, after calculating ADF efficiency statistics and collapsing to the yearly-ticker level.

Table A1. Example of Data After Collapsing to Ticker-Year Level for ADF Tests

Year	Ticker	ADF Efficiency	Broadband	Cellular	Telephone	ETF	ln(GDP)
2010	ARGT	-	9.756	138.886	162.886	1	9.236
2010	BSESN	1.971	0.884	60.630	63.451	0	7.206
2010	BUX	2.233	21.639	120.277	150.077	0	9.489
2010	BVSP	6.748	7.280	100.294	122.094	0	9.342
2010	ECH	6.821	10.415	116.749	136.849	1	9.444
2010	EGPT	3.330	1.627	80.985	91.785	1	7.806
2010	EIDO	3.169	0.926	86.589	103.189	1	8.028
2010	EPHE	3.732		87.862	91.323	1	7.679
2010	EPI	5.266	0.884	60.630	63.451	1	7.206
2010	EPOL	3.875	15.379	121.646	141.746	1	9.439
2010	EPU	5.250	3.206	99.609	110.509	1	8.532
2010	EWM	6.586	7.321	117.902	134.002	1	9.094
2010	EWV	6.561	9.314	81.206	98.706	1	9.183
2010	EWZ	6.183	7.280	100.294	122.094	1	9.342

Appendix B: Including Time Fixed Effects

Table B1. Runs Test: Entity Fixed Effects (5-Year Averages)

	(1)	(2)	(3)	(4)	(5)	(6)
	BBND	CEL	TEL	BBND	CEL	TEL
Broadband	0.009**			0.000		
	(0.036)			(0.999)		
Cellular		0.002**			0.000	
		(0.010)			(0.816)	
Telephones			0.002**			0.000
			(0.012)			(0.738)
lnGDP				0.417**	0.428*	0.411*
				(0.015)	(0.067)	(0.065)
Entity FE	Yes	Yes	Yes	Yes	Yes	Yes
Period FE	No	No	No	No	No	No
N	165	178	178	165	178	178

Table B2. ADF Test: Entity Fixed Effects (5-Year Averages)

	(1)	(2)	(3)	(4)	(5)	(6)
	BBND	CEL	TEL	BBND	CEL	TEL
Broadband	-0.010			-0.029*		
	(0.367)			(0.064)		
Cellular		0.002			0.001	
		(0.273)			(0.734)	
Telephones			0.002			0.002
			(0.249)			(0.659)
lnGDP				0.423*	0.052	0.022
				(0.057)	(0.905)	(0.958)
Entity FE	Yes	Yes	Yes	Yes	Yes	Yes
Period FE	No	No	No	No	No	No
N	165	178	178	165	178	178

Table B3. Variance Ratio: Entity Fixed Effects (5-Year Averages)

	(1)	(2)	(3)	(4)	(5)	(6)
	BBND	CEL	TEL	BBND	CEL	TEL
Broadband	-0.073			-0.022		
	(0.335)			(0.488)		
Cellular		-0.007			0.017	
		(0.326)			(0.305)	
Telephones			-0.005			0.020
			(0.324)			(0.298)
lnGDP				-2.351	-5.600	-6.064
				(0.321)	(0.297)	(0.292)
Entity FE	Yes	Yes	Yes	Yes	Yes	Yes

Table B3. Variance Ratio: Entity Fixed Effects (5-Year Averages)

Period FE	No	No	No	No	No	No
N	165	178	178	165	178	178

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